

GCP Data Analytics Pipeline: NBA Player Performance Analytics

CPSC 482 – Data-Intensive Systems | Project Proposal

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1. Dataset Description

This project will use the **NBA Stats dataset** sourced from the official NBA statistics API (stats.nba.com) via the *nba_api* Python library, supplemented by historical data available on Kaggle. The combined dataset covers player-level box score statistics, team game logs, and advanced metrics spanning the 2000–01 through 2025–26 NBA seasons.

The dataset characteristics are as follows:

Attribute	Details
Source	NBA Stats API (<i>nba_api</i>) + Kaggle historical archives
Approximate Size	~500 MB across all tables (2M+ individual stat rows)
Format	JSON (API responses), CSV (Kaggle exports)
Structure	Structured, relational (player, team, game, season dimensions)
Key Tables	Player game logs, team standings, shot charts, advanced stats
Data Quality	Generally clean; some missing advanced metrics in older seasons; requires deduplication across sources

Dataset Sources:

- **NBA Stats API (*nba_api*)** – Primary data source for current and historical player/team statistics.
https://github.com/swar/nba_api
- **NBA Database (1947–Present)** – Comprehensive Kaggle dataset with historical box scores and player data.
<https://www.kaggle.com/datasets/eoinamoore/historical-nba-data-and-player-box-scores>
- **NBA Database (Wyatt Walsh)** – Large relational SQLite-based dataset with multiple joined tables.
<https://www.kaggle.com/datasets/wyattowalsh/basketball>
- **NBA Players & Team Data** – Clean player and team statistics for supplemental reference.
<https://www.kaggle.com/datasets/loganlauton/nba-players-and-team-data>

2. GCP Services

The pipeline will leverage the following Google Cloud Platform services to ingest, store, process, warehouse, and visualize the data:

Pipeline Stage	GCP Service	Purpose
Data Ingestion	Cloud Functions	Scheduled Python functions to pull data from the NBA API in batch intervals
Raw Data Storage	Cloud Storage (GCS)	Landing zone for raw JSON/CSV files, organized by season and data type
Data Processing	Dataproc (PySpark)	Clean, deduplicate, join, and transform raw data into analytics-ready tables
Data Warehouse	BigQuery	Store processed fact and dimension tables; serve as the query engine for analytics
Analytics / ML	BigQuery ML	Build regression and clustering models directly in BigQuery using SQL
Visualization	Looker Studio	Interactive dashboard connected to BigQuery for exploring player trends

3. Final Product

The final deliverable will be an **interactive Looker Studio dashboard** backed by BigQuery, with a **BigQuery ML regression model** for predictive insights. Key dashboard views include a player performance explorer (scoring, rebounding, and assist trends filterable by player, season, and team), a season comparison tool for tracking career trajectories, and advanced metrics panels covering PER, TS%, BPM, and VORP with league-wide percentile rankings. The BigQuery ML model will forecast points per game based on usage rate, minutes, and historical performance.

4. Justification for a Large-Scale Pipeline

The dataset spans 25+ seasons and over two million rows across multiple related tables. Computing rolling aggregates, cross-season joins, and ML feature engineering at this scale benefits from distributed processing rather than local execution. The project also integrates heterogeneous sources—live JSON from the NBA API and historical CSVs from Kaggle—requiring a structured ingestion and staging layer. BigQuery’s serverless query engine and built-in ML capabilities allow the entire workflow, from ad-hoc analytics to model training, to run without provisioning separate infrastructure.